

How Operating System Boots Up

What Happens When You Turn On Your Computer?

1. PC On

You press the power button.

Electricity starts going to all parts of the computer.

Example:

Like switching on a fan — electricity reaches the motor first.

2. CPU looks for Firmware (BIOS/UEFI)

As soon as power reaches the CPU, the CPU wakes up and looks for a special program called **firmware**.

This firmware is stored on the **BIOS chip** on the motherboard.

- BIOS = Basic Input Output System
- It helps the computer start at the most basic level.

In modern computers → BIOS is replaced by UEFI

UEFI = Unified Extensible Firmware Interface

It is a modern, faster, more secure version of BIOS.

Example:

Think of BIOS/UEFI like the "starter" of a bike.

Before the bike runs, the starter must kick everything into motion.

3. BIOS loads configuration settings

The BIOS/UEFI reads saved settings from a special memory (CMOS).

These settings include:

- Time & Date
- Boot order
- Connected hard disks
- CPU details
- Fan settings etc.

Example:

Like a shopkeeper opening the shop and first checking all items, tools, and lights are in place.

4. BIOS performs POST

POST = **Power-On Self Test**

This is a basic test of all major hardware:

- RAM
- Keyboard
- CPU
- Hard Disk
- Graphics card

If something is missing or faulty, BIOS shows an error (beep sounds or screen error).

Example:

Like a car doing a quick check when you start it — fuel? brakes? engine?

If something is wrong, the car shows a warning light.

5. BIOS/UEFI finds the Bootloader

Now BIOS/UEFI hands over the responsibility to the **bootloader**, which will load the operating system.

BIOS looks for **MBR (Master Boot Record)**

- This is the first sector of the hard disk.
- Contains a tiny program → **bootloader**.

UEFI looks for **EFI System Partition (ESP)**

- Contains .efi files like bootx64.efi.

Example:

Think of BIOS/UEFI like a receptionist.

It guides you to the correct department (bootloader), which will further take you to your cabin (operating system).

6. Bootloader loads the Operating System

The bootloader is a small program whose job is to load the OS.

Examples of bootloaders:

- Windows → **Windows Boot Manager (bootmgr.exe)**

- Linux → **GRUB**
- Mac → **boot.efi**

Once the bootloader starts → it loads:

- Kernel
- Drivers
- System files
- Login screen

Example:

The bootloader is like the lift operator who takes you from the ground floor to your office floor (OS).

Real-Life Example

Suppose you turn on your laptop:

1. Power button pressed → electricity flows.
2. CPU wakes up and says, “Where is the BIOS/UEFI?”
Finds it on the motherboard.
3. BIOS reads saved settings (time, disk details).
4. BIOS checks if RAM, keyboard, and disk are fine (POST).
5. BIOS searches the hard disk for the bootloader.
6. It finds Windows Boot Manager.
7. Boot Manager loads Windows.
8. You see “Starting Windows” → login screen.